

Final Group Project Paper

Introduction

Our core question is “What helps a college basketball team be successful and how does that translate to their players being drafted into the NBA.” To answer this, we are using three tables: a college basketball dataset, an NBA players dataset (which includes player performance and partial draft information), and a draft dataset that includes international players and complete draft data. The idea behind using these tables is to connect team-level success in college to individual player outcomes in the NBA. Instead of just looking at one side (either teams or players), we wanted to see the full pipeline — how team performance, player exposure, and draft outcomes all tie together.

This project focuses on three connected areas:

- College basketball team success
- NBA player performance and draft position
- International players as another pathway into the NBA

Motivation

The motivation for this project came from our interest in basketball and the connection between college basketball and the NBA. As international students, we noticed that college sports have a much larger culture in the United States than what we were used to, especially in basketball. Since we both enjoy the sport, we wanted to better understand how college basketball connects to the professional level. We were also interested in the NBA draft because it plays a major role in how teams build for the future. As fans of the Golden State Warriors, we have seen how quickly teams can rise, fall, and rebuild, which made the draft process even more interesting to us. We also noticed that many important NBA players today come from international backgrounds, which made us want to compare the college route with the

international route. Overall, this project allowed us to use our interest in basketball to study team success, player development, draft value, and different paths into the NBA.

Data

We found most of our datasets on Kaggle because it is one of the easiest and most accessible platforms for getting structured data. We also briefly explored government and official sports data sources, but those were either too limited or not in a usable format for this project. Many of those sources either did not include enough historical data or did not combine team and player statistics in a way that would help us answer our question. Kaggle datasets gave us a good balance of completeness and usability, as they contained mostly clean and organized statistics across multiple years. However, we noticed issues in the nba dataset's draft-related data, especially with missing or incorrect information for international players. To fix this, we used data from Wikipedia to cross-reference and rebuild a more complete draft dataset that includes international players properly.

We chose these particular datasets because they contain a large amount of relevant information about both teams and players. The college basketball dataset includes detailed team-level metrics such as offensive efficiency (ADJOE), defensive efficiency (ADJDE), turnover rate (TOR), shooting efficiency (EFG_O), and postseason performance. These are useful because they go beyond just wins and losses and give insight into how a team plays. For example, two teams might have similar win totals but very different styles, and these metrics help capture that difference. On the other side, the NBA dataset and draft dataset provide player-level information such as draft position, performance stats (points, rebounds, assists), and background information like college or nationality. By combining these, we are able to look at not only which teams succeed, but also whether that success leads to players being drafted and how those players perform in the NBA, including those who did not go through the college system.

The three main datasets used were:

- College basketball dataset: used to study team success, efficiency, seeding, and postseason results.
- NBA player dataset: used to connect players to their college background, draft number, and NBA performance.
- International draft dataset: used to account for players who did not come through the NCAA system.

EDA

During our exploratory data analysis (EDA), we noticed several issues with missing or NULL values, but most of them made sense once we understood the structure of the data. In the college basketball dataset, columns like SEED and POSTSEASON often contain NULL values. This is expected because not every team qualifies for the NCAA tournament, so they would not have a seed or postseason result. In a way, these NULL values carry meaning — they indicate that a team did not reach a certain level of success. In the NBA dataset, we initially noticed that many players had values listed as “None” for college. At first, this appeared to be missing data, but after further analysis, we realized that many of these players were international players who did not go through the NCAA system. Instead of treating this as missing data, we decided to improve the dataset by using a draft dataset from Wikipedia to properly account for international players.

Another thing we noticed is that some variables behave very differently from others. For example, metrics like turnover rate (TOR) and shooting efficiency (EFG_O) tend to show clearer differences between strong and weak teams, while other metrics like three-point shooting percentage (3P_O) are more tightly clustered across teams. This suggests that not all statistics are

equally useful in explaining team success. Some variables provide stronger signals, while others may not have as much variation or impact.

The main data challenges were:

- NULL values in `SEED` and `POSTSEASON`
- Players with `"None"` listed for college
- Missing or incomplete draft information
- International players not being fully represented in the original NBA dataset
- Some variables being more useful than others for explaining success

Questions answered

This data allows us to answer several meaningful questions. The main question is our core one: What helps a College basketball team be successful and how does that translate to their players being drafted into the NBA? Beyond that, we can also explore questions such as how important seeding is in college basketball. Higher-performing teams tend to receive better seeds in tournaments, which usually means they face weaker opponents in the early rounds. This gives them a structural advantage and increases their chances of advancing further. Because of this, seeding is not just a result of success but also something that can reinforce future success within a tournament.

We can also examine which statistical factors are most strongly associated with winning. For example, teams with lower turnover rates tend to perform better because they retain possession more often, while teams with higher shooting efficiency are able to convert their opportunities into points more effectively. Another question we can explore is whether these same factors influence how many players from a team get drafted. It is possible that teams with strong offensive systems or balanced play produce players who are more appealing to NBA scouts. At the same time, we can look at draft

position and see whether players from stronger college teams tend to be drafted earlier compared to players from weaker teams.

The data helps answer questions such as:

- What statistics are most connected to college basketball success?
- How important are tournament seed and postseason performance?
- Do successful college teams produce more drafted players?
- Do players from stronger college teams get drafted earlier?
- How do college players compare to international players?

Raised questions

At the same time, the data raises additional questions about how different pathways to the NBA compare. While we were able to analyze both college and international players, it still raises the question of how scouting evaluates players across completely different systems. For example, even though international players are often drafted slightly later, our analysis shows that their performance is very similar, and in some cases even better, than college players. So, draft position may not always fully reflect a player's potential. It also raises the question of whether NBA teams may undervalue certain types of players during the draft, especially those coming from non-traditional pathways.

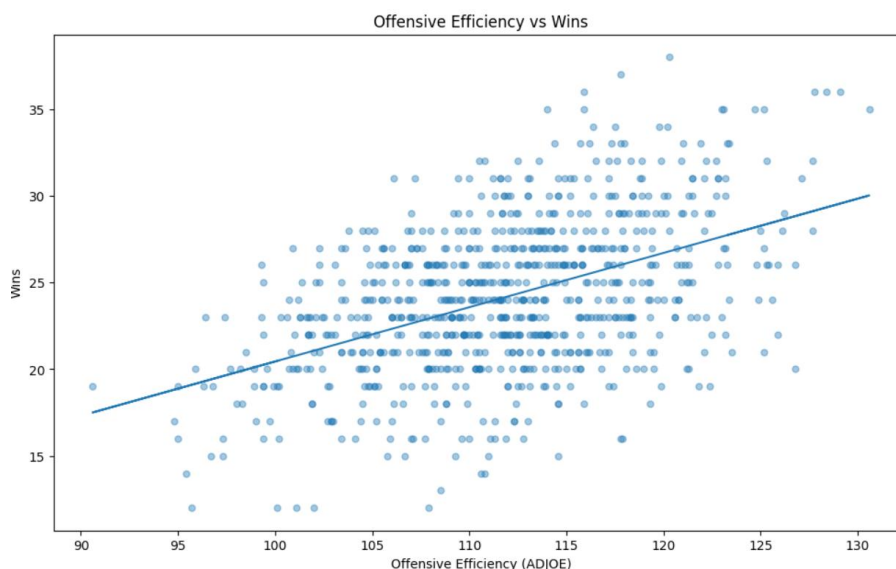
Another question is how well college team success predicts individual player success at the NBA level. While we initially expected a strong connection, our analysis shows that the relationship is more limited. Players from stronger college teams are drafted slightly earlier, but the difference is not very large, and individual performance in the NBA varies significantly. So, individual ability plays a bigger role in determining long-term success than does the level of education. Overall, the results show that both college

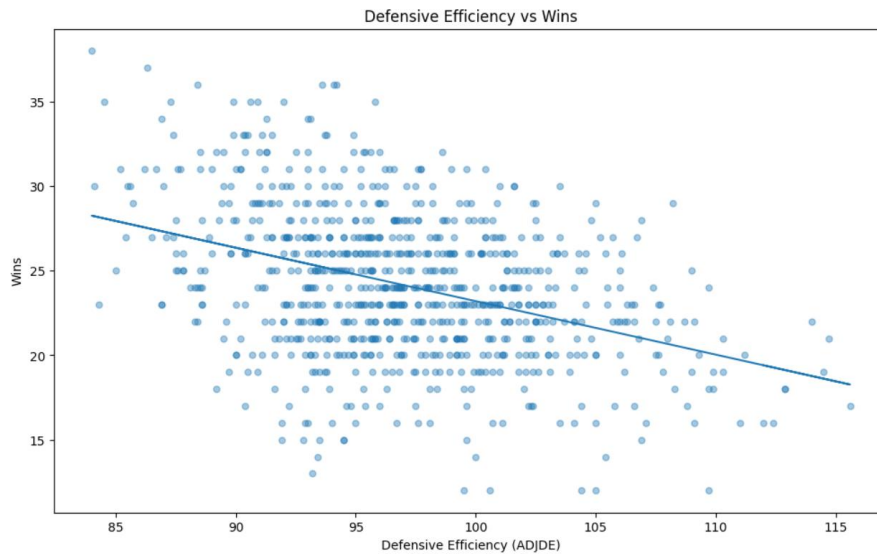
systems and international pathways can produce successful NBA players, and no single path guarantees success.

Plots, Trend & Analysis

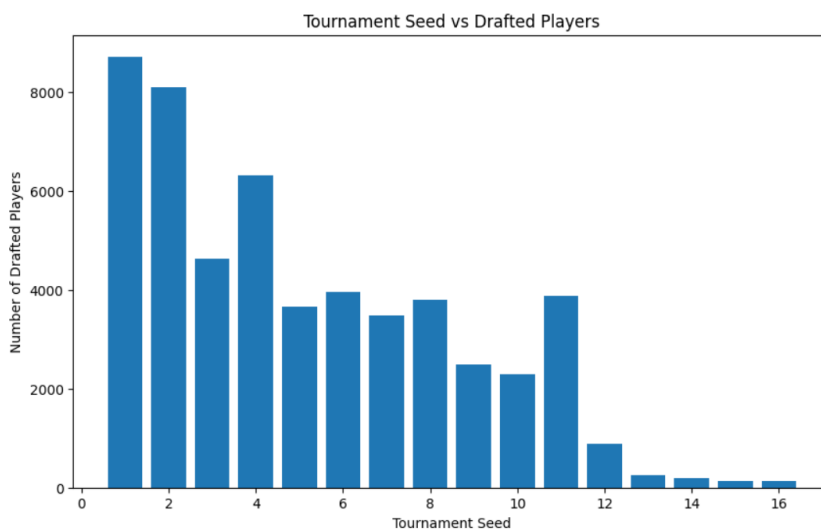
The visualizations also helped support the patterns we saw in SQL. Here are some of the graphs with explanations (rest of them, at the end after the conclusion):

In the offensive efficiency versus wins graph, there is a clear upward trend. As offensive efficiency increases, teams generally appear to win more games. This makes sense because teams that score more efficiently are able to turn possessions into points more often, which gives them a better chance to win over a full season. The defensive efficiency graph shows the opposite pattern. Since a lower ADJDE value is better, the teams with stronger defensive efficiency tend to appear closer to the higher-win area. Together, these graphs show that team success is not based on one single factor. Successful teams usually need both efficient offense and strong defense, though strong offense is more important than strong defense (as shown in SQL code).

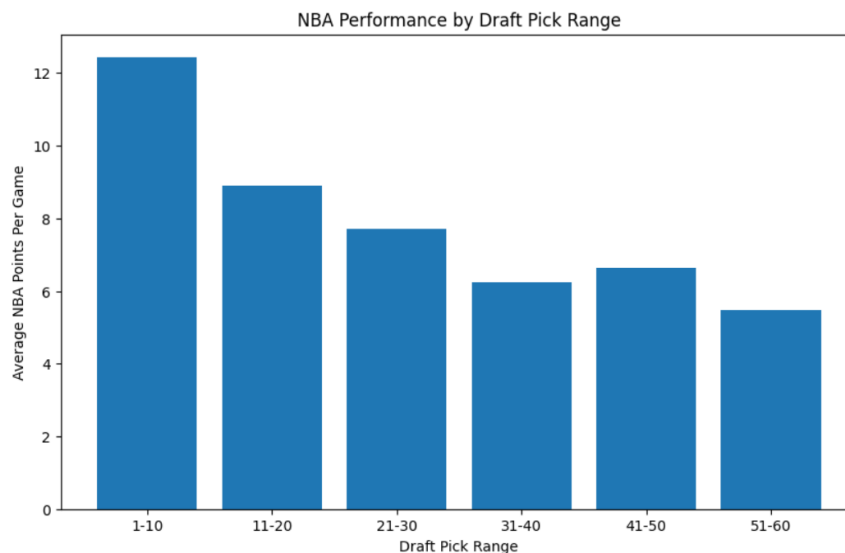
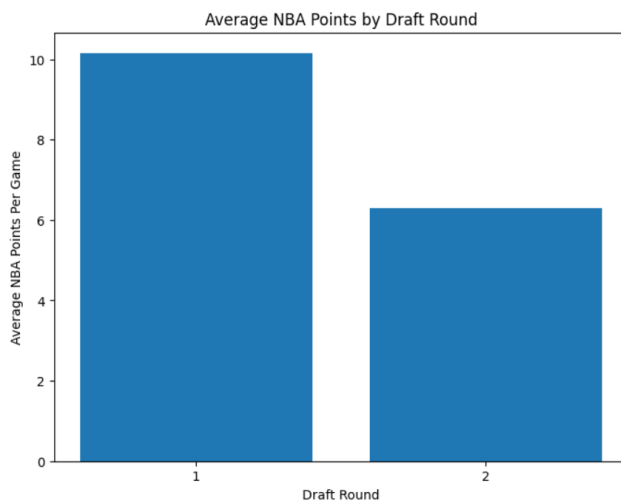




The tournament seed visualization also adds an important part to the analysis. The graph shows that lower-numbered seeds, especially the highest seeds, are connected to more drafted players. This suggests that stronger tournament teams often have more NBA-level talent. This does not mean every player on a top seed will become a strong NBA player, but it does show that team success and draft visibility are connected. A high seed usually comes from strong regular season performance, and that gives players more national attention during the tournament. From a draft perspective, this matters because scouts may be more likely to evaluate players who are competing on successful and highly visible teams.

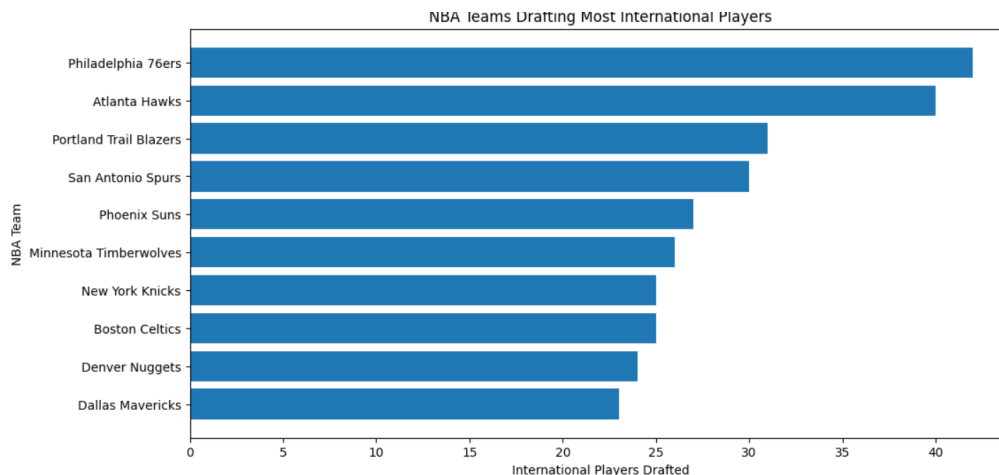


The draft pick versus NBA points per game graphs show why draft position matters. Majority get selected in first draft round and within the rounds, the earlier picks often have higher average points scored per game. Many of the higher-scoring NBA players are found near the lower draft numbers (player is good), meaning they were selected earlier. This supports the idea that NBA teams are usually better at identifying strong talent near the top of the draft (player gets selected with low draft number which means they are a good player).



The Philadelphia 76ers had the highest number of international draft picks, followed by the Atlanta Hawks, Portland Trail Blazers, San Antonio Spurs, and Phoenix Suns. This suggests that some NBA teams may have been

more willing to scout and draft players from outside the traditional college basketball system. This is important for our core question because it shows that college basketball is not the only path to the NBA. While college team success can help players gain visibility, international players can also enter the league through a completely different route. The graph supports the idea that NBA teams evaluate talent from multiple systems, not just NCAA basketball.



The main visualization trends were:

- Offensive efficiency generally increases with wins.
- Lower defensive efficiency is connected to stronger teams.
- Lower-numbered tournament seeds have more drafted players.
- Earlier draft picks usually have higher NBA points per game.
- International players show that college basketball is not the only path to the NBA.

Conclusion

Overall, the trends from the visualizations match the SQL results (since they were made using the SQL queries in python). College success is strongly connected to team-level efficiency, especially offensive efficiency, defensive

efficiency, shooting efficiency, and limiting turnovers. These factors help explain why certain teams win more games and receive better tournament positions. However, when this is connected to NBA draft and performance data, the relationship becomes less direct. Successful college teams and postseason teams tend to produce slightly better draft outcomes and slightly better NBA production in some areas, but the differences are not large enough to say that college team success guarantees NBA success. Team success helps create visibility and context for a player, while the player's individual statistics and NBA performance still need to be included in the analysis.

Because of this, our answer to the core question is not one-sided. What helps a college basketball team succeed is a combination of efficient offense, strong defense, smart possessions, and overall team quality. These traits can help players get noticed by NBA scouts, especially when the team reaches the postseason or earns a strong tournament seed. This supports one of the main ideas from the SQL analysis: college team success gives useful context, but it cannot replace individual player evaluation. A player can benefit from being on a successful team because it gives them exposure and competitive experience, but their NBA production still depends on their own skill and development.

The main findings were:

- College success is strongly connected to efficiency, defense, shooting, and limiting turnovers.
- Strong college teams can give players more visibility.
- Postseason success and strong seeding can help draft attention.
- Draft position matters, but it does not guarantee NBA success.
- Individual player performance still matters more than team success alone.
- International players show that there are multiple paths to the NBA.

Remaining Graphs

